Optimization in Decision Analysis (MOD500) (Course Assignment)

Malvin Varpe - mavar3685@uis.no — Nicolai Rohde Garder - 283419@uis.no

Project Title: A Case Study of Multi-Period Renewable Energy and Battery Scheduling

1 Introduction

In this project, you will study a two-stage stochastic optimization problem for day-ahead operation of a small power system that integrates wind generation and a battery energy storage system. The operator must plan the base energy schedule before knowing actual conditions for the next day and later adjust these decisions when the real wind and demand outcomes are revealed. This setting represents a realistic battery scheduling problem, where the battery acts as a flexible resource to balance supply and demand, mitigate uncertainty in renewable generation, and reduce dependence on costly market purchases. Efficient battery scheduling ensures that renewable energy is utilized effectively, curtailment is minimized, and system reliability is maintained. However, the presence of multiple scenarios for uncertain wind and demand leads to a large-scale optimization problem that is computationally challenging to solve directly. To address this, the project introduces the Benders' decomposition method, a powerful approach that separates the problem into a master problem (day-ahead planning) and several subproblems (real-time adjustments for each scenario). This decomposition allows you to understand how complex stochastic energy scheduling problems can be solved efficiently by exploiting their two-stage structure, a concept widely used in modern energy system optimization.

1.1 Notation

The meaning of the symbols and notation are given below, which are used in the next Section (2) to define a two-stage stochastic optimization model for multi-period renewable energy and battery scheduling.

- MW: Mega Watt,
- MWh: Mega Watt hour,
- T: Multi-period planning horizon,

- $|.|$: this is called mod, and it always gives positive value. For example $|-5| = 5$ and also $|5| = 5$. Remember, we use this symbol in the Convergence checking of Benders' decomposition algorithm.

- $\forall$: this symbol means "for all or for each",

- $\in$: this symbol means "the member of or belongs to". For example, if a set A is given as $A = \{1, 2, 3\}$, then "$a \in A$" means a is the member of the set A, and it could be $a = 1$, $a = 2$, or $a = 3$.

2 Project Description

A power operator runs a small power system for "tomorrow", which is split into several time periods (for example, every hour). Thus the operator plans over a horizon of T time periods ($t = 1, 2, \ldots, T$) covering tomorrow. The power operator has

- a wind farm (free energy but varies a lot), and

- a battery (stores energy like a big rechargeable battery).

Tomorrow's demand for electricity and the amount of wind available are uncertain. That is represented by the uncertainty with three possibilities (scenarios): low, medium, or high need of electricity. The power operator's job is to make a plan today and then adjust it tomorrow once operator knows which scenario actually happens. What the power operator actually decides.

Before knowing future (today, same for all scenarios): For each time period t, operator decides a base amount of energy to provide: x_t (MWh). Think of this as "day-ahead plan".

After knowing what happens (tomorrow, can differ by scenario and time):

- How much to charge the battery (in case excess energy left then save excess energy in the battery): $q_{s,t}$ (MWh)

- How much to discharge the battery (to supply electricity for fulfilling the demand): $r_{s,t}$ (MWh)

- How much wind power is unused (called curtailment): $u_{s,t}$ (MWh).

- How much extra energy to buy from the market (to complete the demand): $m_{s,t}$ (MWh).

- What is the battery power level at the end of the period (state of charge): $S_{s,t}$ (MW).

- Unmet power demand in every scenario s and time t: $l_{s,t}$ (MWh).

(In every notation, subscript s means in every scenario: low/medium/high, and subscript t means in every time period).

Goal: Minimize the total expected cost. Paying the cost

- a cost of day-ahead planed energy.
- a price for any market purchases, if needed
- a penalty, if operator curtail wind energy instead of using it.
- a penalty, if any amount of electricity demand is unserved.
- a cost of charging and discharging battery.

Because operator does not know which scenario will happen, he/she minimizes the average cost across the three scenarios (weighted by their probabilities).

Rules must follow:

- Battery cannot be charged or discharged faster than its limit.
- The battery level (state of charge) must always stay between 0 and its maximum size.
- Battery power level carries over time.
- Meet demand in every period.
- Finish the day reasonably. That means
  1. Operator must end the last period with the battery neither empty nor unrealistically full (stay within a reasonable final range).
  2. This prevents "cheating" by using the whole battery at the last minute and leaving it empty for the next day.

2.1 Data/Known Parameters

- $T = 24$: Tomorrow's 24 hour planning horizon.
- Scenario set $s = \{1, 2, 3\}$ with probabilities $\pi_1 = 0.3,\ \pi_2 = 0.4, \pi_3 = 0.3$.
- Day-ahead base schedule cost per MWh energy: c_t for each period t is given in the Table (2) (see, Appendix Section (4)).
- A big penalty for unserved energy demand: $P_{\text{unmet}} = 100000$ (NOK/MWh).
- For each period t and scenario s:
  1. Net load $D_{s,t}$ (MWh) is given in Table (3) (see, Appendix Section (4)).

2. Available wind energy $W_{s,t}$ (MWh) is given in Table (4) (see, Appendix Section (4)).
3. Real-time market purchase price $p_{s,t}$ (NOK/MWh) is given in Table (5) (see, Appendix Section (4)).
4. Wind energy curtailment penalty $k_{s,t} = 320$ (NOK/MWh) in every scenario and hour.

- Hourly maximum market purchase capacity $M_{\text{cap},t}$ (MWh) is given in the Table (6) (see, Appendix Section (4)).
- Battery limits: energy capacity $E_{\max} = 120$ (MWh), charge and discharge power limits $P_{\text{max-charge}} = 45$ (MW), $P_{\text{max-discharge}} = 55$ (MW), respectively, efficiencies of charging and discharging $\eta_{\text{charge}} = 0.95$ and $\eta_{\text{discharge}} = 0.95$.
- Initial state of charge of battery $S_0 = 60$ (MWh), desired minimum and maximum limit of state of charge of battery are $S_{\min} = 55$ (MWh), $S_{\max} = 85$ (MWh), respectively.
- Optional battery throughput cost $\beta = 12$ (NOK/MWh) applied to $(q_{s,t} + r_{s,t})$ (charging/discharging battery).

2.2 Two-Stage Stochastic Optimization Model of Multi-Period Renewable Energy and Battery Scheduling

The scenario tree is displayed in Figure (1),
where scenario $s = \{s_1, s_2, s_3\} = \{\text{low, medium, high demand}\}$ with probabilities $\pi_s = \{\pi_1, \pi_2, \pi_3\} = \{0.3, 0.4, 0.3\}$, respectively.

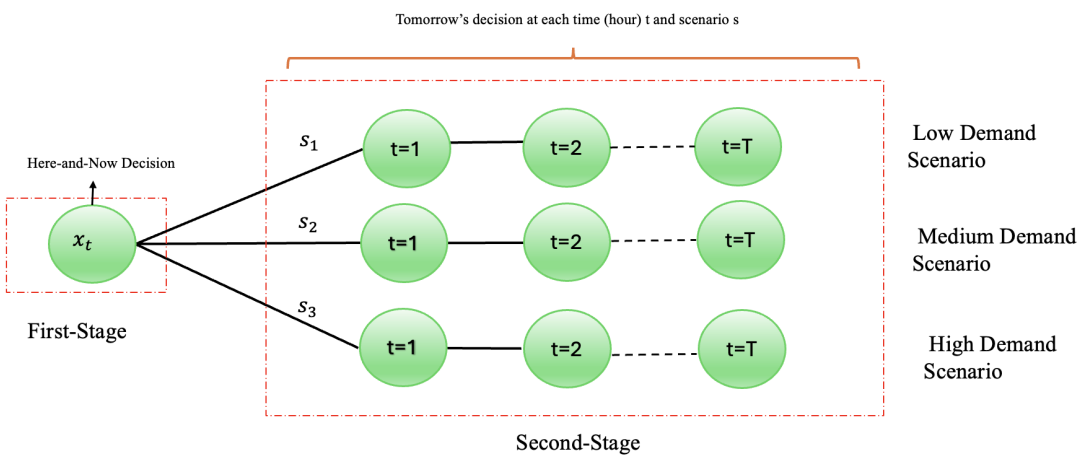

Figure 1: Scenario Tree of Two-Stage Multi-Period Stochastic Optimization Model

Cost of day ahead planned energy: $\sum_{t=1}^{24} c_t x_t$

Tomorrow's total expected cost:

$$\sum_{s=1}^{3}\sum_{t=1}^{24}\pi_s[\underbrace{p_{s,t}m_{s,t}}_{\text{purchasing cost}} + \underbrace{k_{s,t}u_{s,t}}_{\text{penality on wind power curtailment}} + \underbrace{P_{\text{unmet}}l_{s,t}}_{\text{penality on unmet demand}} + \underbrace{\beta(q_{s,t}+r_{s,t})}_{\text{charging/discharging battery cost}}].$$

Constraints:

1. Non-negativity Condition

$$x_t, q_{s,t}, r_{s,t}, u_{s,t}, m_{s,t}, S_{s,t}, l_{s,t} \geq 0, \text{ for all } t, s.$$

2. Natural capacity on x_t (don't schedule more than high-demand)

$$0 \leq x_t \leq X_{\max,t}, \quad \text{for all } t \in T.$$

   where $X_{\max,t} = \max_{s\in S} D_{s,t}$, for all $t \in T$.

3. Energy purchased from market limit. The market purchase constraint ensures that the operator cannot buy more energy from the market than a predefined maximum market capacity.

$$m_{s,t} \leq M_{\text{cap},t}, \text{ for all } s = \{1,2,3\}, t \in T.$$

4. Battery power limits (in every scenario s and time period t):

$$q_{s,t} \leq P_{\text{max-charge}},$$

$$r_{s,t} \leq P_{\text{max-discharge}}.$$

5. Battery energy bounds (in every scenario s and time period t):

$$0 \leq S_{s,t} \leq E_{\max}.$$

6. Initial state of charge dynamics (time-coupled and in every scenario s and time period t):

$$S_{s,1} = S_0 + \eta_{\text{charge}} q_{s,1} - \eta_{\text{discharge}} r_{s,1},$$

$$S_{s,t} = S_{s,t-1} + \eta_{\text{charge}} q_{s,t} - \eta_{\text{discharge}} r_{s,t}, \text{ for } t = 2, 3, \ldots T.$$

7. Terminal state of charge condition (in every scenario s):

$$S_{\min} \leq S_{s,T} \leq S_{\max}.$$

   That means in the end of period battery should not left empty or over charged.

8. Supply-demand balance (stage-linking, in every scenario s and time period t):

$$x_t + r_{s,t} + m_{s,t} + (W_{s,t} - u_{s,t}) + l_{s,t} = D_{s,t} + q_{s,t}.$$

Mathematical Formulation

$$\min_{x_t, q_{s,t}, r_{s,t}, m_{s,t}, u_{s,t}, l_{s,t}, S_{s,t}} \underbrace{\sum_{t=1}^{24} c_t x_t}_{\text{First-Stage}} + \underbrace{\sum_{s=1}^{3} \sum_{t=1}^{24} \pi_s [p_{s,t} m_{s,t} + k_{s,t} u_{s,t} + P_{\text{unmet}} l_{s,t} + \beta(q_{s,t} + r_{s,t})]}_{\text{Second-Stage}}$$

subject to

$$x_t, q_{s,t}, r_{s,t}, u_{s,t}, m_{s,t}, l_{s,t}, S_{s,t} \geq 0, \text{ for all } t, s$$

$$0 \leq x_t \leq X_{\text{max},t}, \text{ for all } t$$

$$m_{s,t} \leq M_{\text{cap},t}, \text{ for all } t, s$$

$$q_{s,t} \leq P_{\text{max-charge}}, \text{ for all } t, s$$

$$r_{s,t} \leq P_{\text{max-discharge}}, \text{ for all } t, s$$

$$0 \leq S_{s,t} \leq E_{\text{max}}, \text{ for all } t, s$$

$$S_{s,1} = S_0 + \eta_{\text{charge}} q_{s,1} - \eta_{\text{discharge}} r_{s,1}$$

$$S_{s,t} = S_{s,t-1} + \eta_{\text{charge}} q_{s,t} - \eta_{\text{discharge}} r_{s,t}, \text{ for } t = 2, 3, \ldots T.$$

$$S_{\min} \leq S_{s,T} \leq S_{\max}, \text{ for all } s$$

$$\underbrace{x_t + r_{s,t} + m_{s,t} + (W_{s,t} - u_{s,t}) + l_{s,t} = D_{s,t} + q_{s,t}, \text{ for all } t, s.}_{\text{Stage-Linking Constraint}} \tag{1}$$

3 Required Tasks

The following tasks are your assignments. **It is not recommended to use any different symbols or notation other than given in this project. If in case you need to use any different notation or symbols in any following task, please define the meaning of that notation/symbols clearly.**

3.1 Task 1

1. Define the types of all the decision variables, such as continuous variable (fractional values), integer variable, in the two-stage stochastic optimization model (1).

   Solution:

   Types of Decision Variables

   The types of all decision variables in the two-stage stochastic optimization model are as follows:

- x_t **(MWh):** Amount of base energy scheduled in period t (day-ahead plan).
  Type: Continuous.
- $q_{s,t}$ **(MWh):** Amount of energy charged to the battery in scenario s, period t.
  Type: Continuous.
- $r_{s,t}$ **(MWh):** Amount of energy discharged from the battery in scenario s, period t.
  Type: Continuous.
- $u_{s,t}$ **(MWh):** Amount of wind energy curtailed (unused) in scenario s, period t.
  Type: Continuous.
- $m_{s,t}$ **(MWh):** Amount of energy purchased from the market in scenario s, period t.
  Type: Continuous.
- $S_{s,t}$ **(MWh):** State of charge of the battery at the end of period t in scenario s.
  Type: Continuous.
- $l_{s,t}$ **(MWh):** Amount of unmet demand in scenario s, period t.
  Type: Continuous.

All variables are continuous. There are no integers of the decision variables in this model.

2. Write clearly two-stage framing for decision variables. For example, which decision variable belongs to first-stage and which belongs to second-stage?

Solution:

Variable	**Stage**	**Description**
x_t	First-stage	Day-ahead base energy schedule
$q_{s,t}$	Second-stage	Battery charging (per scenario)
$r_{s,t}$	Second-stage	Battery discharging (per scenario)
$u_{s,t}$	Second-stage	Wind curtailment (per scenario)
$m_{s,t}$	Second-stage	Market purchase (per scenario)
$S_{s,t}$	Second-stage	Battery state of charge (per scenario)
$l_{s,t}$	Second-stage	Unmet demand (per scenario)

Table 1: Decision variables and stage classification.

3.2 Task 2

Write the Master problem for the two-stage stochastic optimization model (1).

Solution: The master problem determines the optimal first-stage decisions x_t and an auxiliary variable α that represents the expected second-stage cost. Initially, α is unconstrained except for a lower bound, and Benders' cuts are added iteratively to approximate the true recourse cost.

Master Problem Formulation

$$
\begin{aligned}
\min_{x_t,\alpha} \quad & \sum_{t=1}^{T} c_t x_t + \alpha \\
\text{s.t.} \quad & 0 \le x_t \le X_{\max,t}, \quad \forall t = 1, \dots, T, \\
& \alpha \ge \alpha_{\text{down}}, \quad \text{(initial lower bound, e.g., } \alpha_{\text{down}} = 0\text{)}, \\
& \alpha \ge \text{Benders' cut k}, \quad \forall k \in \text{previous iterations.}
\end{aligned}
$$

Where:

- x_t: Day-ahead energy schedule for period t (first-stage decision).
- α: Auxiliary variable representing the expected second-stage cost.

Interpretation

- The objective minimizes the sum of day-ahead cost and the expected subproblem cost.
- The auxiliary variable α is constrained from below by Benders' cuts, which approximate the second-state function.
- Each new iteration adds a cut, tightening the approximation and guiding x_t towards the optimal solution.

3.3 Task 3

Write the sub-problem(s) (per scenario s) for the two-stage stochastic optimization model (1).

Solution: Given the first-stage decision x_t from the master problem, the subproblem for each scenario $s \in \{1, 2, 3\}$ minimizes the second-stage (recourse) cost subject to operational constraints. The formulation for scenario s is:

$$
\begin{aligned}
\min_{q_{s,t}, r_{s,t}, m_{s,t}, u_{s,t}, l_{s,t}, S_{s,t}} \quad & \sum_{t=1}^{24} \Big[p_{s,t} m_{s,t} + k_{s,t} u_{s,t} + P_{\text{unmet}} l_{s,t} + \beta(q_{s,t} + r_{s,t}) \Big] \\
\text{s.t.} \quad & q_{s,t}, r_{s,t}, m_{s,t}, u_{s,t}, l_{s,t}, S_{s,t} \geq 0, \quad \forall t \\
& m_{s,t} \leq M_{\text{cap},t}, \quad \forall t \\
& q_{s,t} \leq P_{\text{max-charge}}, \quad r_{s,t} \leq P_{\text{max-discharge}}, \quad \forall t \\
& 0 \leq S_{s,t} \leq E_{\max}, \quad \forall t \\
& S_{s,1} = S_0 + \eta_{\text{charge}} q_{s,1} - \frac{1}{\eta_{\text{discharge}}} r_{s,1} \\
& S_{s,t} = S_{s,t-1} + \eta_{\text{charge}} q_{s,t} - \frac{1}{\eta_{\text{discharge}}} r_{s,t}, \quad t = 2, \ldots, 24 \\
& S_{\min} \leq S_{s,24} \leq S_{\max}, \\
& x_t + r_{s,t} + m_{s,t} + (W_{s,t} - u_{s,t}) + l_{s,t} = D_{s,t} + q_{s,t}, \quad \forall t
\end{aligned}
$$

This subproblem is solved for each scenario s given the fixed first-stage decision x_t. NOTE: As the battery discharge efficiency is 0.95, it does not make sense to use this as a factor when multiplying with the discharged amount. In practise, this would mean that the battery is losing less energy than the system gets. Therefore, I have changed the equation to use $1/\eta_{\text{discharge}}$ to account for the inefficiency of the battery. If the system extracts r energy from the battery, the battery is drained $\frac{1}{\eta_{\text{discharge}}} r$.

3.4 Task 4

Write the Benders' cut for the two-stage stochastic optimization model (1), and clearly mention the dual variable.

Solution: After solving the subproblem for each scenario s at iteration k, we obtain some new variables. To make the functions easier to write, we are introducing three more symbols:

- $\hat{\alpha}_s^{(k)}$: Optimal second-stage cost for scenario s given $x^{(k)}$.
- $\lambda_{s,t}^{(k)}$: Dual variable (shadow price) associated with the supply-demand balance constraint in scenario s, period t. This value is how much the optimal solution changes when changing the right hand side of the constraint by one.
- $g_{s,t}^{(k)} = -\lambda_{s,t}^{(k)}$: Subgradient at $x^{(k)}$ for each scenario at time t. This is how much the optimal cost is changing per x at x^k. If increasing the left hand side by one (if increasing x by one, the LHS is increased), the RHS needs to drop by one to be satisfied.

The Benders' cut added to the master problem is then:

$$\alpha \geq \sum_{s \in S} \pi_s \Big(\hat{\alpha}_s^{(k)} + \sum_{t=1}^{T} g_{s,t}^{(k)} (x_t - x_t^{(k)}) \Big).$$

- The term $\hat{\alpha}_s^{(k)}$ represents the constant part of the recourse cost for scenario s at the current solution.
- The summation involving $\lambda_{s,t}^{(k)}(x_t - x_t^{(k)})$ adjusts the cut based on how the first-stage decision changes relative to the previous iteration.
- Each new cut refines the lower bound on α, making the master problem approximate the second-stage cost more accurately.

3.5 Task 5

Write the Step 1 and Step 2 of Benders' decomposition algorithm for the two-stage stochastic optimization model (1).

Solution: Step 1: Initialize the Master Problem

1. Set iteration counter $k = 0$.
2. Initialize the set of Benders' cuts.
3. Define the initial master problem:

$$\min_{x_t, \alpha} \quad \sum_{t=1}^{T} c_t x_t + \alpha$$

subject to:

$$0 \leq x_t \leq X_{\max,t}, \quad \forall t = 1, \ldots, T,$$

and

$$\alpha \geq \alpha_{\text{down}}, \quad \text{(initial lower bound, we use } \alpha_{\text{down}} = 0\text{)}.$$

4. Solve the master problem to obtain $x_t^{(k)}$ and $\alpha^{(k)}$.

Step 2: Solve Subproblems for Each Scenario

1. For each scenario $s \in \{1, 2, 3\}$, solve the second-stage subproblem:

$$\min_{q_{s,t}, r_{s,t}, m_{s,t}, u_{s,t}, l_{s,t}, S_{s,t}} \sum_{t=1}^{T} \Big[p_{s,t} m_{s,t} + k_{s,t} u_{s,t} + P_{\text{unmet}} l_{s,t} + \beta (q_{s,t} + r_{s,t}) \Big]$$

subject to all operational constraints and given $x_t = x_t^{(k)}$.

2. Compute the optimal second-stage cost:

$$\hat{\alpha}_s^{(k)} = \text{optimal objective value for scenario } s.$$

3. Extract the dual variables:

$$\lambda_{s,t}^{(k)} = \text{dual value of the supply-demand balance constraint for scenario } s, \text{ period } t.$$

4. Calculate the expected cost:

$$\bar{\alpha}^{(k)} = \sum_{s \in S} \pi_s \hat{\alpha}_s^{(k)}.$$

3.6 Task 6

Write the Step 3 and Step 4 of Benders' decomposition algorithm for the two-stage stochastic optimization model (1).

Solution: Step 3: Add Benders' Cut to the Master Problem

1. Using the results from Step 2, construct the Benders' cut:

$$\alpha \geq \sum_{s \in S} \pi_s \Big(\hat{\alpha}_s^{(k)} + \sum_{t=1}^{T} g_{s,t}^{(k)} (x_t - x_t^{(k)}) \Big).$$

2. Add this cut to the master problem. The master problem now includes all cuts from previous iterations:

$$\alpha \geq \sum_{s \in S} \pi_s \Big(\hat{\alpha}_s^{(j)} + \sum_{t=1}^{T} g_{s,t}^{(j)} (x_t - x_t^{(j)}) \Big), \quad \forall j = 1, \ldots, k.$$

Step 4: Check Convergence

1. Compute the actual expected second-stage cost for the current solution:

$$\bar{\alpha}^{(k)} = \sum_{s \in S} \pi_s \hat{\alpha}_s^{(k)}.$$

2. Compare $\bar{\alpha}^{(k)}$ with the master problem's $\alpha^{(k)}$:

$$\text{If } |\bar{\alpha}^{(k)} - \alpha^{(k)}| \leq \epsilon, \text{ then stop.}$$

3. Otherwise, increment $k := k + 1$ and return to Step 1.

Step 3 refines the master problem by adding a new cut that improves the approximation of the second-state cost function. Step 4 ensures the algorithm stops when the master problem's lower bound ($\alpha^{(k)}$) is close enough to the true expected cost ($\bar{\alpha}^{(k)}$).

3.7 Task 7

Implement the Benders' decomposition algorithm for solving the two-stage stochastic optimization model (1) in Python using Pyomo library and GLPK solver.

Solution: Attach the python code file separately with proper documentation for each step of Benders' decomposition algorithm, and also attach the CSV files of optimal values of all decision variables.

3.8 Task 8

Plot the convergence graph of Benders' decomposition algorithm for solving the two-stage stochastic optimization model (1).

Solution: Here we can see the graph from the results that we obtained. From the graph we can see that the model is decreasing the upper bound, and lower bound, so that the absolute gap is decreasing. After around 30 iterations this value is already very low, and decreasing slowly after, until it reaches 100 iterations. This means that the model might not have hit the absolute error value we wanted, but it looks like it is good enough for a solution.

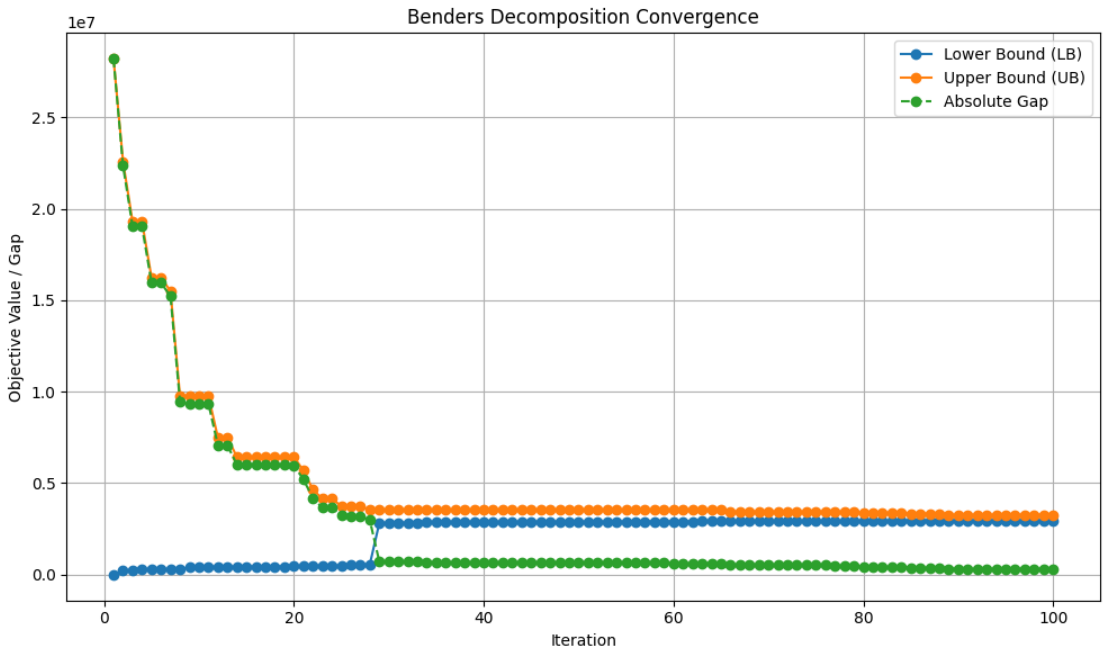

Figure 2: Convergence graph for benders solver

4 Appendix

All the tables of data are given below:

Table 2: Day-ahead base schedule cost per MWh energy for each hour

Hour (t)	Cost c_t (NOK/MWh)
1	760
2	740
3	720
4	710
5	720
6	770
7	830
8	900
9	940
10	920
11	900
12	880
13	870
14	860
15	860
16	880
17	930
18	990
19	1010
20	990
21	950
22	900
23	860
24	820

Table 3: Hourly demand profiles $D_{s,t}$ (MWh) under three scenarios

Hour (t)	Low demand $s = 1$	Medium demand $s = 2$	High demand $s = 3$
1	266.0	280	302.4
2	256.5	270	291.6
3	247.0	260	280.8
4	242.3	255	275.4
5	247.0	260	280.8
6	275.5	290	313.2
7	313.5	330	356.4
8	342.0	360	388.8
9	361.0	380	410.4
10	351.5	370	399.6
11	346.8	365	394.2
12	337.3	355	383.4
13	332.5	350	378.0
14	327.8	345	372.6
15	332.5	350	378.0
16	351.5	370	399.6
17	380.0	400	432.0
18	399.0	420	453.6
19	389.5	410	442.8
20	370.5	390	421.2
21	342.0	360	388.8
22	313.5	330	356.4
23	294.5	310	334.8
24	275.5	290	313.2

Table 4: Hourly available wind energy $W_{s,t}$ (MWh) under three scenarios

Hour (t)	Low demand $s = 1$	Medium demand $s = 2$	High demand $s = 3$
1	93.1	118.8	148.5
2	103.0	131.7	164.6
3	124.7	159.0	199.1
4	141.7	181.6	227.0
5	165.8	212.0	265.0
6	185.4	236.9	296.1
7	207.8	265.8	332.3
8	225.2	287.7	359.6
9	207.2	264.0	330.1
10	184.1	234.9	293.7
11	161.4	206.9	258.6
12	147.8	189.3	236.6
13	136.6	174.7	218.4
14	128.8	164.0	205.0
15	120.1	153.9	192.4
16	130.8	168.7	210.9
17	146.8	189.4	236.7
18	163.1	210.7	263.4
19	177.7	229.5	287.0
20	169.0	218.4	273.0
21	146.8	189.5	236.8
22	121.5	156.3	195.4
23	103.4	133.1	166.4
24	97.3	125.2	156.5

Table 5: Hourly real-time market purchase price $p_{s,t}$ (NOK/MWh) under three scenarios

Hour (t)	Low demand $s = 1$	Medium demand $s = 2$	High demand $s = 3$
1	837.0	900	990.0
2	818.4	880	968.0
3	799.8	860	946.0
4	790.5	850	935.0
5	799.8	860	946.0
6	837.0	900	990.0
7	911.4	980	1078.0
8	967.2	1040	1144.0
9	1023.0	1100	1210.0
10	1004.4	1080	1188.0
11	985.8	1060	1166.0
12	967.2	1040	1144.0
13	950.6	1020	1122.0
14	930.0	1000	1100.0
15	930.0	1000	1100.0
16	950.6	1020	1122.0
17	1004.4	1080	1188.0
18	1078.8	1160	1276.0
19	1097.4	1180	1298.0
20	1078.8	1160	1276.0
21	1023.0	1100	1210.0
22	967.2	1040	1144.0
23	911.4	980	1078.0
24	874.2	940	1034.0

Table 6: Hourly maximum market purchase capacity $M_{\text{cap},t}$ (MWh)

Hour (t)	$M_{\text{cap},t}$ (MWh)
1	140
2	140
3	140
4	140
5	150
6	170
7	190
8	200
9	200
10	190
11	180
12	170
13	160
14	160
15	160
16	170
17	152
18	168
19	168
20	160
21	180
22	160
23	150
24	145